

November-December 2025 Examination
I B. Tech. (4YDC) EXAM
ME-10008: Overview of Mechanical Engineering & Graphics

Time : 3 Hrs]

TOTAL NO. OF QUESTIONS IN THIS PAPER: 5

[Max Marks : 70

Note : All questions are compulsory. Assume suitable data if necessary. Attempt any two parts from question 1 to 4 in answer book. Only Question 5 is to be attempted in drawing sheet.

Q No.	Question	Marks	CO	BL
1.	(a) Define elasticity, plasticity, and brittleness.	7	1	1
	(b) A material has modulus of rigidity equal to $0.4 \times 10^5 \text{ N/mm}^2$ and bulk modulus equal to $0.8 \times 10^5 \text{ N/mm}^2$. Find its Young's Modulus and Poisson's Ratio	7		2
	(c) A steel platform supported by four legs of square cross section of $60 \times 60 \text{ mm}^2$. A load of 2000 kN rests on the platform which is transferred on all four legs equally. Determine stress induced in the legs.	7		2
2.	(a) Define completely constrained, incompletely constrained and successfully constrained motion.	7	2	1
	(b) In single slider crank mechanism, length of the connecting rod is 160 mm and ratio of connecting rod to crank is 4. The crank rotates at 6000 RPM. Determine 1. Maximum angular displacement of connecting rod, 2. Displacement of slider when crank angle is 80° .	7		2
	(c) Classify gears on the basis of different criteria.	7		1
3.	(a) Define the term density, viscosity, and specific gravity.	7	3	1
	(b) Describe Steady flow, Uniform flow, and Laminar flow.	7		1
	(c) Explain Lagrangian and Eulerian method of fluid flow study.	7		1
4.	(a) Explain working of 4 stroke petrol engine with the help of necessary diagrams.	7	4	1
	(b) Explain the terminology bore, stroke, brake horse power and mechanical efficiency in context with I C Engine.	7		1
	(c) Write full form of ABS, EBD, and ESC in context of active safety system of automobile with their function.	7		1
5.	(a) Draw isometric view from front and top view as shown in figure 1.	14	5	3
OR				
	(b) Draw front view, top view and side view from pictorial view shown in figure 2.	14		3

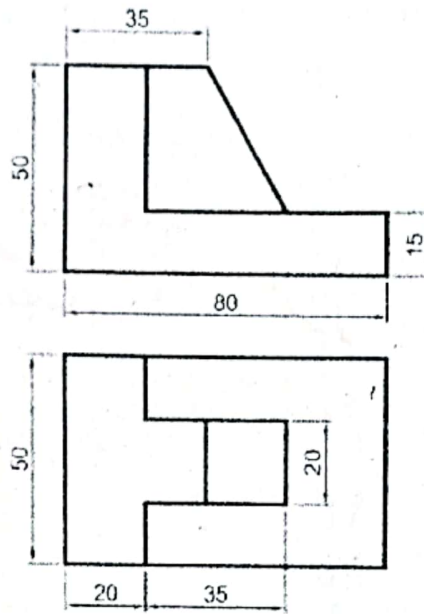


Figure 1

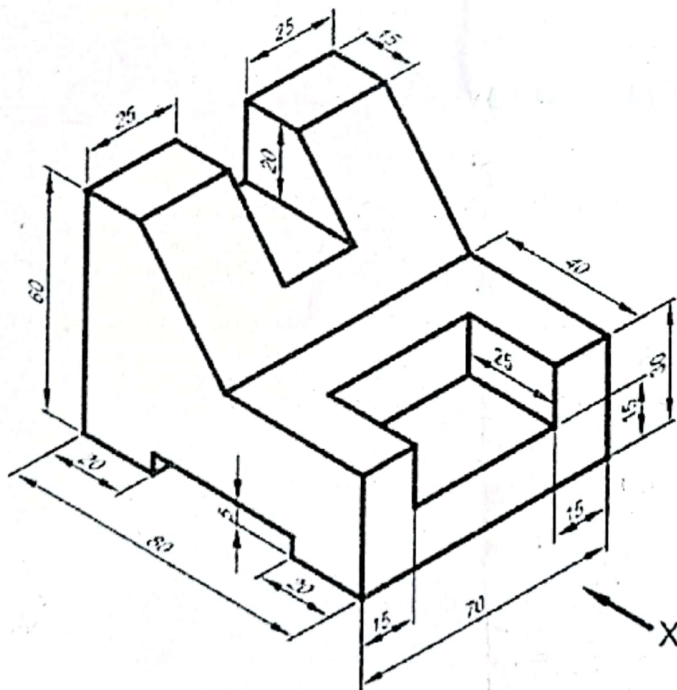


Figure 2

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